Tabayyun-Cert

Certified, Corroboration-Gated Quarantine for Poisoned Test-Time Memory in Federated, Agentic AI-Native Networks

Principal Investigator
Raja Abraheem Rashid Ejaz — Department of Computer Science, Brunel University London; Telecommunications Research Lab (TRL), IBA Karachi

Partner institution
To be confirmed (collaborating university)

Research area
Security of LLM memory · Federated learning · AI-native 6G / Open RAN · Certified robustness

One-sentence thesis

A test-time memory that learns from unvetted sources can be made *provably* unable to let any minority of sources change what an AI-native network does, by writing every unvetted report into a fast, decaying, per-source quarantine and promoting it to shared memory only after independent corroboration, with a closed-form certificate on how far the trusted memory and the downstream network action can be moved.

Headline contributions

1. **Threat model.** The first formal threat model for *parametric* test-time memory (Titans / Hope / ATLAS-style) shared across federated clients and agentic O-RAN controllers. It is mapped to OWASP ASI06 and MITRE ATLAS AML.T0080.000.
2. **Mechanism.** A two-tier *source-conditional* memory: a per-source quarantine with a fast forget gate, and a promoted tier that is momentum-free and corroboration-gated, federated with a two-level k-of-n trimmed-mean rule.
3. **Theory.** Three closed-form results, each with a stated failure regime:
   - Sybil-limited promotion;
   - a drift certificate that depends only on the forget gate and the honest spread;
   - certified invariance of RAN actions, with a benign-promotion delay law.
4. **Audited evaluation.** Adaptive attacks, a pre-registered negative regime (strong non-IID), and Colosseum O-RAN trace replays.

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1 Executive Summary

Novelty confidence (honest assessment)

Medium-high. Searches to 8 Nov 2026 found:

- attacks on federated test-time adaptation (FedPoisonTTP), on SSM hidden states (HiSPA) and on retrieval memory (MINJA, AgentPoison, GhostWriter);
- an empirical HiSPA filter (CLASP);
- a claim-level, Islamic-provenance router over *external* knowledge (ISNAD);
- a federated Titans architecture with *no* security analysis (FedNL, arXiv 2605.16350).

None of these gives a certified defence for gradient-written, federated test-time memory. IEEE Xplore, ACM DL and OpenReview must be re-queried before submission (Section 10).

Large language models are acquiring *memory that learns at inference time*. Titans [1] updates a neural long-term memory at test time by gradient descent on an associative loss, driven by a momentary-plus-momentum "surprise" signal and a data-dependent forget gate. Miras [2] recasts every modern recurrent model as a choice of memory architecture, attentional bias, retention gate and memory optimiser. Nested Learning and Hope [4] organise such memories into a Continuum Memory System (CMS) of levels updated at different frequencies. FedNL [5] has already proposed training Titans-style self-referential memories federatedly across clients. In parallel, operators are deploying LLM-based copilots and *agentic* xApps/rApps inside Open RAN [6, 7], and 3GPP Release 20 has opened the first 6G study items with AI-native air-interface and "common AI enabler" work.

The security community has formalised the resulting risk:

- **OWASP** lists *Memory & Context Poisoning* as ASI06 in its Top 10 for Agentic Applications (December 2025) [8].
- **MITRE ATLAS** added *AI Agent Context Poisoning: Memory* (AML.T0080.000) [9].
- **Microsoft** documented real-world "AI Recommendation Poisoning" that targets assistant memory [10].

Measured attacks on *retrieval-store* memories reach injection success near 95–98% [11, 12, 13]. Every published defence, however, is heuristic, and every published attack and defence targets either retrieval stores or single-model hidden states [15, 16]. **No one has shown how to let a federated, gradient-written memory learn from unvetted sources while provably bounding what a minority of sources can make the network do.**

Tabayyun-Cert closes this gap with three ingredients:

1. **Quarantine.** Every write from source s lands in a per-source quarantine memory M_s^Q with a fast forget gate. It is readable only by the same source and only for low-risk queries.
2. **Corroboration-gated promotion.** Periodically, only directions in key space on which at least k independent sources agree are promoted, via a coordinate-wise trimmed mean, into a *momentum-free* promoted memory M^P. Only M^P is federated, with the same k-of-n rule applied across clients.
3. **Certificates.** Closed-form bounds on the drift of M^P under f Byzantine sources, and on whether any downstream RAN action can flip. Queries that cannot be certified are routed to abstention or fallback, exactly as in the PI's NTC risk-controlled controller.

The design principle, *verify a consequential report from an unvetted source before acting on it, and*

seek corroboration from independent sources, is motivated by Qur'an 49:6 and its classical exegesis (Section 3). This is conceptual motivation only. Every claim in this proposal is carried by the theorems and the pre-registered experiments.

2 Problem Statement and Stakes

2.1 From retrieval memory to parametric test-time memory

Most current agent memories are external stores: vector databases, RAG indices and conversation logs. Attacks on them are now mature:

- **MINJA** [11] injects malicious records through query-only interaction.
- **AgentPoison** [12] reaches high attack success at poison rates below 0.1%.
- **GhostWriter** [13] reports near-universal injection ($\approx$98%) and $\approx$60% activation.
- **Bad Memory** [14] frames persistent memory compromise as a "stored XSS" of agents.

The next generation of models moves memory *into parameters that are updated at inference time*. For these memories:

1. A single adversarial context chunk changes weights. In SSMs, HiSPA [15] shows that short trigger phrases irreversibly overwrite hidden state, causing large performance drops.
2. Titans' momentum term carries an adversarial write forward in time. The *persistence* of poison is therefore a function of the gate parameters.
3. Once memories are federated (FedNL [5]), poison written at one site propagates to every site.

2.2 Why AI-native networks make this urgent

In an AI-native RAN, an agentic controller with test-time memory reads telemetry, tickets, xApp reports and operator instructions, and writes what it learns. Consider a compromised xApp, a spoofed E2 report or a poisoned trouble ticket that writes the association "cell 17 interference $\Rightarrow$ reduce transmit power". That association persists across sessions, and through federation across sites. The harm is concrete: coverage holes, SLA violations for URLLC slices, or denial of service by misconfiguration. O-RAN WG11 maintains a threat-modelling report [18], and DRL xApps have been shown to be poisonable [19]. No standard, however, yet addresses the *test-time memory* of RAN agents.

Research question

Can a federated test-time memory keep learning from unvetted sources, including tools, telemetry, tickets, other agents and other sites, while guaranteeing in closed form that no coalition of at most f sources can (i) move the trusted memory beyond a computable radius, or (ii) flip a certified network action?

3 Inspiration Statement

Qur'an 49:6 (al-Ḥujurāt)

يَا أَيُّهَا الَّذِينَ آمَنُوا إِن جَاءَكُمْ فَاسِقٌ بِنَبَإٍ فَتَبَيَّنُوا أَن تُصِيبُوا قَوْمًا بِجَهَالَةٍ فَتُصْبِحُوا عَلَىٰ مَا فَعَلْتُمْ نَادِمِينَ

"O you who have believed, if there comes to you a disobedient one with information, investigate, lest you harm a people out of ignorance and become, over what you have done, regretful." (Sahih International)

Key terms.

- *fāsiq*: an unreliable or untrustworthy transmitter.
- *naba'*: news of consequence.
- *fa-tabayyanū*: "seek clarity, verify". A well-known variant reading is *fa-tathabbatū*, "be certain, deliberate".

Classical exegesis (as consulted).

- **Ibn Kathīr:** Allah ordered investigating the news that the wicked bring, to make sure of its authenticity. He reports the occasion of revelation linked to al-Walīd b. 'Uqba's report about Banū al-Muṣṭaliq. Modern editors note weaknesses in some chains for this occasion, so this proposal relies on the verse's *general rule*, not on the incident.
- **al-Ṭabarī:** he records both readings as well known and close in meaning, and glosses the instruction as أمهلوا حتى تعرفوا صحته، لا تعجلوا بقبوله ("wait until you know its soundness; do not hasten to accept it").
- **al-Jaṣṣāṣ (Aḥkām al-Qur'ān):** as reported in secondary sources, such a report is not acted upon unless it is investigated through other sources and confirmed.

Supplementary: Qur'an 83:14 and the hadith of the heart's stain

كَلَّا بَلْ رَانَ عَلَىٰ قُلُوبِهِم مَّا كَانُوا يَكْسِبُونَ

"No! Rather, the stain has covered their hearts of that which they were earning." (Sahih International)

The hadith in *Jāmi' al-Tirmidhī* 3334 (graded ḥasan ṣaḥīḥ by al-Tirmidhī) says that a sin leaves a black spot on the heart. The heart is polished clean if the person refrains and repents, but the spot increases if the sin is repeated, until it covers the heart.

Mapping from principle to mechanism. Table 1 separates *design principles*, which the mechanism actually implements, from *analogies*, which are illustrative only.

Recommended wording for the paper (one paragraph, after the technical introduction)

"Our design principle, *verify a consequential report from an unvetted source before acting on it, and confirm it through independent sources*, is motivated by Qur'an 49:6 and its classical exegesis. We use this purely as conceptual motivation, in the same way neuroscience motivates memory architectures; all claims are established by the theorems and experiments that follow."

Source concept	Mechanism in Tabayyun-Cert	Status
Verify a consequential report from an unreliable transmitter before acting (49:6)	Writes from unvetted sources go to a per-source quarantine M_s^Q and cannot drive high-risk actions	Design principle
Confirmation through other sources (al-Jaṣṣāṣ)	k-of-n corroboration gate on promotion to M^P	Design principle
"Do not hasten to accept it" (al-Ṭabarī)	Promotion only at period τ, after evidence accumulates	Design principle
A stain that is polished away early but compounds if repeated (83:14; Tirmidhī 3334)	Momentum accumulation of poison; the quarantine's fast forget gate as "polishing"; no momentum in the trusted tier	Analogy
Avoid general suspicion of all reports (modern commentary)	Source-conditional verification: verification cost is spent only where provenance is weak	Design principle

Table 1: Concept-to-mechanism mapping. No claim is made that the text describes algorithms.

4 Threat Model

System. A federation of N clients (operator sites, edge clouds or near-RT RIC instances). Each client hosts an agentic model with a test-time memory. The memory ingests chunks x_t from n *sources* $s(t) \in \{1, \ldots, n\}$: xApp/rApp reports, E2 telemetry streams, tool outputs, operator tickets and other agents. Each source is authenticated, for example through E2/A1 credentials, so the source identity is known even when its content is untrusted.

Adversary.

- **Capability.** Controls at most f sources per client and at most f_c clients. It can make arbitrary writes through controlled sources, including coordinated, adaptive and low-norm repeated writes (the "rayn attack", Section 9.3).
- **Knowledge.** Knows the architecture, the gate parameters, k, τ and the trimming level (white-box with respect to the defence), but not the honest sources' data.
- **Goals.**
  (a) *Targeted insertion*: make $M^P(q^\star) \approx v^\star$ for a chosen key/value.
  (b) *Amnesia*: erase a benign association (HiSPA-style).
  (c) *Action flipping*: change the RAN action chosen for a target query.
  (d) *Degradation*: reduce benign recall.

Out of scope, and stated as failure modes.

- Forgery of source identity (Sybil identities beyond f).
- A shared poisoned upstream feed that makes "independent" sources correlated. This is a supply-chain attack, OWASP ASI04.
- Compromise of the aggregation server.
- Jailbreaks of the frozen backbone.

Each of these is discussed in the risk analysis rather than ignored.

5 Formal Setup

We follow the Titans/Miras notation [1, 2]. For input x_t, keys and values are $k_t = x_t W_K \in \mathbb{R}^{d_k}$ and $v_t = x_t W_V \in \mathbb{R}^{d_v}$, and the query is $q_t = x_t W_Q$. The memory $M : \mathbb{R}^{d_k} \to \mathbb{R}^{d_v}$ is linear ($M \in \mathbb{R}^{d_v \times d_k}$) for the theory, and an MLP in experiments, where the theory is applied to its local linearisation. The attentional bias is

$$\ell(M; x_t) = \|M(k_t) - v_t\|_2^2 . \tag{1}$$

The Titans update with momentum ("surprise") S_t, learning rate θ_t, momentum η_t and forget gate α_t is

$$S_t = \eta_t S_{t-1} - \theta_t \nabla \ell(M_{t-1}; x_t), \qquad M_t = (1 - \alpha_t) M_{t-1} + S_t. \tag{2}$$

In the Miras taxonomy this is (MLP memory, ℓ_2 attentional bias, weight-decay retention, momentum-GD optimiser).

Lemma (persistence of a single write). Consider one write with gradient g at time 0, $\|\theta g\| \le \theta G$, and constant gates. Its contribution to M_T is $\theta g \sum_{j=0}^{T} (1-\alpha)^{T-j} \eta^j$, whose norm is at most

$$\theta G \, (T+1) \, c^T, \qquad c = \max(\eta, 1-\alpha), \tag{3}$$

and also at most $\theta G / (\alpha(1-\eta))$. The asymptotic half-life of poison is therefore $\approx \ln 2 / (-\ln c)$. *Momentum lengthens persistence.* This formalises the "stain that compounds" intuition, and it motivates the design choice that the trusted tier carries *no* momentum.

6 Method: Tabayyun-Cert

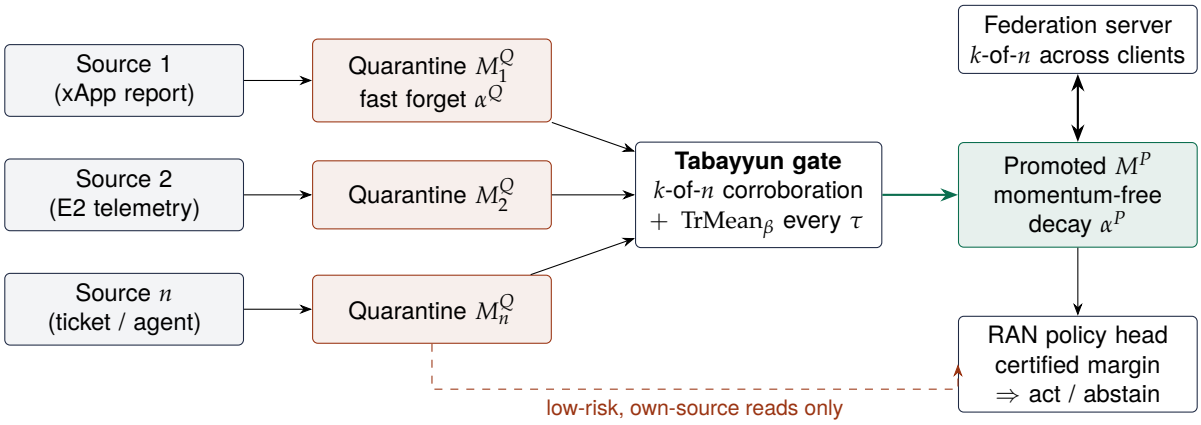


Figure 1: Tabayyun-Cert architecture. Unvetted writes never reach the federated, action-driving memory without independent corroboration.

6.1 Tiers

- **Quarantine tier** $\{M_s^Q\}_{s=1}^n$. There is one memory per source, updated by Eq. (2) with a strong forget gate α^Q and per-write clipping $\|\theta_t \nabla \ell\| \le \theta G$. Momentum is allowed here; it aids learning, and Lemma (3) bounds its persistence.

- **Promoted tier** M^P. It is updated only at promotion boundaries, by a momentum-free rule with constant decay $\alpha^P \ll \alpha^Q$. This is the CMS "slow level" and the only tier that is federated or that can drive high-risk actions.

6.2 Read rule (risk-routed)

$$y_t = M^P(q_t) + \lambda_r \, \mathbf{1}[r(q_t) \le r_0] \; M^Q_{s(t)}(q_t), \tag{4}$$

where $r(q)$ is a risk score: a static action-class map, for example "read-only analytics" versus "reconfigure power or handover". High-risk queries see only corroborated knowledge.

6.3 Promotion rule (the tabayyun gate)

Every τ steps, let $\Delta_s = M^Q_s - M^{Q(\text{prev})}_s$ be the quarantine delta of source s, and let $\mathcal{K} = \{u_1, \dots, u_K\}$ be an orthonormal basis of the active key subspace (top-K right singular vectors of recent keys). For each direction u:

$$\hat{m}_u = \text{median}_s \left(\Delta_s u\right) \quad \text{(coordinate-wise in } \mathbb{R}^{d_v}\text{)}, \tag{5}$$

$$\mathcal{C}_u = \{\, s : \; \cos(\Delta_s u, \hat{m}_u) \ge \gamma \text{ and } \; \|\Delta_s u\| \ge \nu \,\}, \tag{6}$$

$$\Delta^P = \sum_{u:\; |\mathcal{C}_u| \ge k} \text{TrMean}_\beta \left(\{\Delta_s u\}_{s=1}^n\right) u^\top, \tag{7}$$

$$M^P \leftarrow (1 - \alpha^P) M^P + \Delta^P. \tag{8}$$

Quarantine deltas are then decayed or reset. Unpromoted information is never deleted from the *log*: an append-only provenance record (source, time, hash) supports later audit and unlearning (see Proposal 2).

6.4 Federation

Clients transmit only Δ^P. The server applies the same corroboration-and-trimming rule across clients, using k_c-of-N and trimming $\beta_c \ge f_c$. This produces a two-level nested hierarchy: sources within a client, then clients within the federation. It mirrors the parameter-group hierarchy of the PI's Nested-MARL [32] and the per-tier synchronisation of TierFed [34].

6.5 Algorithm

7 Theory

Assumption 1 (Honest spread)**.** In each promotion round, for every direction u and output coordinate, the honest sources' values $(\Delta_s u)_i$ lie in an interval of width at most Δ_H.

Assumption 2 (Minority and trimming)**.** $f < \min(k, \, n/2)$ and the trimmed mean removes $\beta \ge f$ values on each side.

Theorem 1 (Sybil-limited promotion)**.** *Under Assumption 2, a direction u is promoted only if at least $k - f$ honest sources corroborate it. In particular, if no honest source carries a component along u above the threshold ν, the adversary's influence on M^P along u is exactly zero.*

Proof sketch. $|\mathcal{C}_u| \le f + |\mathcal{C}_u \cap \mathcal{H}|$. Promotion requires $|\mathcal{C}_u| \ge k$, so $|\mathcal{C}_u \cap \mathcal{H}| \ge k - f$. If the honest set has no component above ν along u, then $|\mathcal{C}_u \cap \mathcal{H}| = 0 < k - f$ and u is not promoted. □

Algorithm 1: Tabayyun-Cert (one client)

Input: stream $(x_t, s(t))$; gates $\alpha^Q, \alpha^P, \eta, \theta$; clip G; period τ; k, γ, ν, β; risk map r

for $t = 1, 2, \ldots$ **do**
- $s \leftarrow s(t)$; compute k_t, v_t, q_t;
- $g \leftarrow \mathrm{clip}_G(\nabla \ell(M_s^Q; x_t))$; update (S_s, M_s^Q) by Eq. (2) with α^Q;
- $y_t \leftarrow M^P(q_t) + \lambda_r \mathbf{1}[r(q_t) \le r_0] M_s^Q(q_t)$;
- **if** *action*(y_t) *is high-risk* ***and*** *margin* $m(q_t) \le 2L \|q_t\| D_R$ **then**
  - route to abstain / safe fallback (NTC-style);
- **if** $t \bmod \tau = 0$ **then**
  - **foreach** $u \in \mathcal{K}$ **do**
    - compute $\hat{m}_u, \mathcal{C}_u$; **if** $|\mathcal{C}_u| \ge k$ **then** Δ^P += $\mathrm{TrMean}_\beta(\{\Delta_s u\}) u^\top$;
  - $M^P \leftarrow (1 - \alpha^P) M^P + \Delta^P$; decay all M_s^Q; log $(\Delta^P, \mathcal{C}, D_R)$;
  - send Δ^P to server; receive aggregated promoted memory;

Theorem 2 (Closed-form drift certificate)**.** *Let $\tilde{M}_R^P$ be the promoted memory that would result from honest sources alone, under the same promotion decisions, after R promotion rounds. Under Assumptions 1–2 and constant α^P,*

$$\left\| M_R^P - \tilde{M}_R^P \right\|_F \;\le\; D_R := \sqrt{d_v K}\, \Delta_H \, \frac{1 - (1 - \alpha^P)^R}{\alpha^P} \;\le\; \frac{\sqrt{d_v K}\, \Delta_H}{\alpha^P}. \tag{9}$$

Proof sketch. With at least f values trimmed per side, every coordinate of TrMean_β lies in the honest range, and so does the honest-only aggregate. The per-round coordinate error is therefore at most Δ_H, giving at most $\sqrt{d_v K}\, \Delta_H$ in Frobenius norm over the K promoted directions. The error recursion $E_R = (1 - \alpha^P) E_{R-1} + e_R$ unrolls into a geometric sum. Because M^P is momentum-free, no extra factor $1/(1 - \eta)$ appears; with momentum one would get the strictly larger bound $D_R/(1 - \eta)$. □

Theorem 3 (Certified action invariance and benign-promotion delay)**.** (*a*) *Let the policy head h be L-Lipschitz from the memory read to the logits, and let $m(q)$ be the top-1 logit margin at query q. If*

$$m(q) \;>\; 2L \, \|q\| \, D_R, \tag{10}$$

then no admissible adversary can change the action chosen at q.
(*b*) *If each of n_H honest sources independently observes a true association with probability ρ per round, the probability that it is promoted within r rounds is*

$$\mathbb{P}(\textit{promoted by } r) = \mathbb{P}\big(\mathrm{Bin}\big(n_H,\, 1 - (1 - \rho)^r\big) \ge k\big)\,. \tag{11}$$

Proof sketch. (a) $\left\|(M^P - \tilde{M}^P) q\right\| \le \left\|M^P - \tilde{M}^P\right\|_{op} \|q\| \le D_R \|q\|$. Each logit moves by at most $L D_R \|q\|$, so the margin shrinks by at most $2 L D_R \|q\|$. (b) Each honest source has observed the association by round r with probability $1 - (1 - \rho)^r$, independently across sources. □

When the method helps or fails.

- **Helps when.** Theorem 2 is informative when the honest spread is small relative to action margins, i.e. $\Delta_H \ll \alpha^P m(q) / (2L \|q\| \sqrt{d_v K})$.
- **Fails when (vacuous bound).** The bound becomes vacuous under strong source heterogeneity: Dirichlet(≤ 0.1) site splits, or facts unique to one site.

- **Fails when (benign facts never promoted).** Theorem 3(b) predicts that client-unique benign facts are *never* promoted. They remain usable only for low-risk, same-source reads.

Both regimes are **pre-registered as expected failures** and will be reported.

Corollary (cost of security). The expected benign-recall loss for high-risk queries equals the mass of true associations with corroboration probability below the k-threshold. Together with Theorem 3(b), this gives a closed-form operating curve over (k, τ) that a network operator can tune, analogous to the accuracy budget ε in NTC.

8 Research Questions and Pre-Registered Hypotheses

Hypothesis 1 (Persistence)**.** In ungated Titans memory, the half-life of a poisoned association scales as $\ln 2/(-\ln \max(\eta, 1-\alpha))$ (Lemma (3)). A repeated low-norm "rayn" attack achieves the same targeted attack success rate (ASR) as a single large write at a fraction of its per-write norm, and so evades norm-based filters.

Hypothesis 2 (Defence)**.** At matched clean recall (within 1 point on MQAR/BABILong), Tabayyun-Cert reduces targeted ASR by $\geq 50\%$ relative to Titans with FedAvg, and by $\geq 25\%$ relative to the best robust-aggregation baseline, for $f/n \leq 0.2$, *including against the adaptive attacker*.

Hypothesis 3 (Certificate tightness)**.** On IID and mildly non-IID splits, the certified drift D_R is within a factor of 10 of the measured worst-case drift under the strongest adaptive attack, and the certified-action fraction is $\geq 60\%$ on Colosseum replay queries.

Hypothesis 4 (Negative regime, pre-registered)**.** Under Dirichlet(0.1) site heterogeneity, Tabayyun-Cert's high-risk benign recall drops by more than 10 points and the certificate becomes vacuous. Ungated memory wins on utility in this regime.

9 Experimental Plan

9.1 Models

- **Memory models.** Titans-style models (MAC and LMM variants) at 125M and 340M parameters, built on an open reimplementation; Hope/CMS at small scale; and an ATLAS/Omega variant [3] for generality.
- **Retrofitted LLM.** A neural-memory module retrofitted onto a 0.5B open LLM, used for the network-copilot experiments.
- **Federated substrate.** The FedNL [5] three-level setup is the natural federated baseline.

9.2 Datasets and benchmarks

9.3 Attacks (all implemented; the adaptive attack is mandatory)

1. **Single-shot targeted write**: CounterFact-style association with a large norm.
2. **Rayn attack**: repeated low-norm writes timed to exploit momentum.
3. **HiSPA-style amnesia triggers** adapted to gradient memory [15].

Purpose	Data
Memory recall under poisoning	MQAR (Zoology) [20]; BABILong [21]
Targeted false associations	CounterFact [22] (subject–relation–object rewrites injected as context streams)
O-RAN agent decisions	Colosseum / ColO-RAN KPI traces [23] for slicing and scheduling; synthetic beam/CSI memory tasks from DeepMIMO [24] and Sionna [25]
Poisoned telemetry sources	CICIoT2023 [26]; N-BaIoT [27]
Cross-session protocol	Bad Memory evaluation protocol [14]
Network-copilot QA	Telecom QA benchmark (e.g. TeleQnA / GSMA Open-Telco suites)

Table 2: Datasets.

4. **Retrieval-style injection** ported to parametric memory: MINJA/AgentPoison-style bridging writes [11, 12].
5. **Adaptive corroboration-forging attack**: f colluding sources that know $(k, \gamma, \nu, \beta, \tau)$. They (i) place their deltas at the edge of the honest range, (ii) aim to shift the median, and (iii) wait for honest corroboration of a nearby direction and piggy-back on it. The attacker optimises over this space with gradient-based search on a differentiable relaxation of the gate.
6. **Federated attack**: f_c malicious clients transmit crafted Δ^P.

9.4 Baselines

- Titans with FedAvg.
- Titans with Krum [28], coordinate-wise trimmed mean [29], or FLTrust [30] applied to memory deltas.
- The Miras robust-loss variants [2].
- CLASP [16].
- AM-Sentry (GhostWriter defence) [13].
- ISNAD routing [17], adapted.
- OWASP-style TTL expiry of suspicious entries [8].
- Clipping-only, and DP-noise memory (link to Proposal 3).

9.5 Metrics

- **Attack effect.** Targeted ASR; persistence half-life (steps and sessions).
- **Utility.** Benign recall (MQAR accuracy, BABILong score); promotion delay and false-reject rate of true facts.
- **Certificates.** Certified drift D_R against measured drift; certified-action fraction.
- **Network outcome.** RAN KPI regret (throughput, latency, SLA violations) in Colosseum replays.
- **Cost.** Communication per tier.
- **Statistics.** Paired bootstrap CIs over ≥ 5 seeds, and Holm-corrected tests on the two pre-declared primary contrasts. This follows the audited protocol of the PI's NTC and TierFed papers.

9.6 Ablations

- Remove quarantine (single tier).
- Remove corroboration (trimmed mean only).
- Add momentum to M^P, to test Theorem 2's $1/(1-\eta)$ penalty.
- Sweep k, τ, γ, α^Q/α^P.
- Compare a learned trust score with a static source-class trust map.

9.7 Compute budget

- **Pretraining.** 340M parameters on ∼15B tokens, $\approx 6ND \approx 3 \times 10^{19}$ FLOPs, which is ≈30 h on 4×L40S at 40% utilisation. This step can be avoided by retrofitting.
- **Federated and attack sweeps.** 300–500 GPU-hours.
- **Total.** ≈800 L40S-hours, feasible on a single node over 12 months.

10 Novelty Audit

Searches performed (web/arXiv-indexed, up to 8 Nov 2026).

- Memory and context poisoning: "OWASP Agentic ASI06 memory poisoning"; "MITRE ATLAS agent context poisoning".
- Parametric memory: "poisoning test-time memory Titans"; "adversarial attack test-time learning memory Titans Atlas"; "poisoning attack Titans neural long-term memory OR Federated Nested Learning Byzantine".
- Certificates and network agents: "certified robustness test-time training memory poisoning bound"; "O-RAN xApp LLM agent memory poisoning".
- Named neighbours: "FedPoisonTTP"; "HiSPA"; "isnad provenance LLM agents".

Claimed delta

The first source-conditional, two-tier *parametric* test-time memory with (i) k-of-n corroboration-gated promotion across sources and federated clients, (ii) a closed-form drift certificate in terms of Titans' gate parameters, and (iii) certified invariance of downstream network actions, together with an adaptive-attack evaluation and a pre-registered failure regime.

Must re-check before submission

- Re-run the search strings in IEEE Xplore, ACM DL, OpenReview (ICLR 2027 submissions), and the USENIX Security / CCS / S&P / NDSS 2026 programmes.
- Add "federated test-time memory poisoning certified" and "Byzantine Titans".
- Recent 2026 preprints cited here are not peer-reviewed; treat their numbers as provisional.

11 Expected Contributions, Outputs and Impact

Closest work	ID	What it lacks relative to Tabayyun-Cert
FedNL (Federated Nested Learning)	2605.16350	Federated Titans memories; *no* poisoning, Byzantine or certificate analysis. This is our substrate, not a competitor
FedPoisonTTP	2511.19248	Attack only; vision entropy-TTA; no parametric sequence memory, no defence
HiSPA	2601.01972	Attack on Mamba hidden states; single model; no defence
CLASP	2603.12206	Empirical token-level filter; no tiers, no bound
ISNAD (isnād–rijāl)	2607.24117	Claim-level provenance routing over *external* knowledge; no parametric memory, federation or certificate. **Must cite** (same metaphor family)
MINJA / AgentPoison / GhostWriter	2503.03704 / 2407.12784 / 2607.06595	Retrieval-store memories; heuristic defences only
Systematic memory-poisoning study	2606.04329	Taxonomy; no certified defence
Trust or admission control for agent memory	2603.02240 / 2603.04549	Admission scoring; no Byzantine bound, no federation
Dynamic online poisoning certificates	2502.16737	Online certificates for mean estimation and linear classifiers only
Krum / trimmed mean / FLTrust	—	Aggregate gradients of fixed models, not inner-loop memory; no source-conditional tiers

Table 3: Closest prior work and the precise delta.

- **Paper A (security venue: USENIX Security / IEEE S&P / CCS).** "Poison That Learns: Persistence Attacks on Test-Time Memory and a Certified Quarantine Defence." Contents: threat model, rayn and adaptive attacks, Theorems 1–3, and the full evaluation.
- **Paper B (IEEE JSAC / TIFS / TMC).** The O-RAN instantiation, with Colosseum replays and certified RAN actions.
- **Artefacts.** An open benchmark of poisoning attacks on parametric test-time memory, the first of its kind, plus reproducible code with one-command regeneration of all numbers (the PI's standard practice).

Why this can become a reference work.

- **Precedent.** Krum, trimmed mean and randomized smoothing [31] became heavily cited because each paired a simple rule with a provable guarantee at the moment a new threat surface appeared.
- **Timing.** Test-time memory is that surface now: OWASP and MITRE formalised the threat class in 2025, Google publicised Titans-line memories, and FedNL proposed federating them in 2026.
- **Benchmark value.** A first certified defence plus a public attack benchmark is the form that tends to be adopted as a baseline.

12 Risks and Mitigations

Risk	Mitigation
Certificate too loose for deep memory	Report empirical-vs-certified ratios; use per-direction subspace bounds; layer-wise linearisation with measured Lipschitz constants
Reviewers see "just robust aggregation"	The novelty is source-conditional tiering *inside* a recurrent, gradient-written memory, momentum exclusion from the trusted tier, and *action-level* certificates; ablations isolate each
Sybils and correlated upstream feeds	Bind sources to E2/A1 credentials; state the correlated-feed attack as an explicit failure mode and measure it
Benign facts starved of corroboration	Low-risk same-source reads; operating curve over (k, τ); report the starvation rate
Titans code availability	Use open reimplementations and FedNL; Hope/CMS small-scale reproduction

13 Work Plan (12 months)

Months	Work package	Deliverable
1–2	WP1: threat model; attack suite on parametric memory (single-shot, rayn, HiSPA-style, MINJA-ported)	Attack benchmark v0; persistence law (H1)
3–5	WP2: quarantine and corroboration mechanism; Theorems 1–2; MQAR/BABILong	Defence v1; theory note
6–8	WP3: federation (FedNL substrate) and O-RAN agent on Colosseum traces; Theorem 3	Certified-action results
9–10	WP4: adaptive corroboration-forging attacker; negative-regime runs	Robustness report
11–12	WP5: artefact release; Paper A submission; Paper B draft	Submissions, code, benchmark

14 Collaboration Model

- **PI** (Rashid Ejaz): mechanism design, federated/O-RAN systems, the 6G testbed link through TRL, and audited evaluation.
- **Partner group** (proposed): theory review and tightening of certificates for MLP memories; access to larger compute for 1B-scale confirmation; security-venue co-authorship.
- **Shared governance**: a pre-registration document fixing primary contrasts and negative regimes before runs.

15 Fit with the PI's Prior Work

- **Nested-MARL** [32]: multi-timescale parameter groups with an EMA-anchored slow group, mirrored by the promoted tier.
- **NTC** [33]: error-stickiness model and certified, Bonferroni-corrected abstention, reused here for persistence and action certificates.
- **TierFed** [34]: per-tier synchronisation under byte quotas, plus the DP traffic side channel.

- **FedSRIMU**: federated unlearning, connected to the provenance log.

Tabayyun-Cert is the natural convergence of these lines on the PI's stated current focus: poisoning and certified defences for test-time memory in federated learning.

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